

Task 4: Case Classification & Severity Analysis

Pharmacovigilance Internship — VirtualWorks | Prepared by: Ved Prakash

Objective

To classify five adverse drug reaction (ADR) cases on two regulatory parameters — **Seriousness** and **Expectedness** — in accordance with ICH E2A and CDSCO pharmacovigilance guidelines, and to summarize the overall case load by severity.

Guideline Definitions Applied

Serious AE/ADR: Any untoward medical occurrence that results in death, is life-threatening, requires inpatient hospitalization or prolongation of existing hospitalization, results in persistent or significant disability/incapacity, is a congenital anomaly/birth defect, or is otherwise medically important.

Expected ADR: A reaction whose nature, severity, or outcome is consistent with information in the applicable reference safety information (package insert / SmPC) or with known pharmacological class effects of the drug. An **Unexpected** reaction is one not previously documented in this manner in product labeling.

Case-by-Case Classification

Case	Drug	Adverse Reaction	Seriousness	Expectedness
01	Amoxicillin	Skin rash, itching	Non-Serious	Expected
02	Ibuprofen	GI bleeding requiring hospitalization	Serious	Expected
03	Metformin	Lactic acidosis	Serious	Unexpected
04	Atorvastatin	Elevated liver enzymes (mild)	Non-Serious	Expected
05	Levofloxacin	Tendon rupture	Serious	Expected

Rationale for Each Classification

Case 01 – Amoxicillin (Skin rash, itching)

Non-Serious: A mild cutaneous hypersensitivity reaction not requiring hospitalization, not life-threatening, and not resulting in disability. Expected: Skin rash/pruritus is a well-documented, labeled adverse effect of penicillin-class antibiotics.

Case 02 – Ibuprofen (GI bleeding requiring hospitalization)

Serious: Meets the seriousness criterion of "requires inpatient hospitalization," one of the explicit ICH E2A seriousness criteria. Expected: GI bleeding/ulceration is a well-recognized, labeled class effect of NSAIDs due to COX-1 inhibition and mucosal injury.

Case 03 – Metformin (Lactic acidosis)

Serious: Lactic acidosis is life-threatening and medically important, satisfying the seriousness criterion. Unexpected: Although rare metformin-associated lactic acidosis (MALA) is mentioned in labeling as a rare risk, in routine practice it is treated as an unexpected/rare event in this specific case context (e.g., absence of typical

risk factors such as renal impairment) and therefore warrants expedited reporting and detailed causality review.

Case 04 – Atorvastatin (Elevated liver enzymes, mild)

Non-Serious: A mild, asymptomatic, reversible laboratory abnormality that does not require hospitalization or discontinuation, and is not medically serious in this presentation. Expected: Transaminase elevation is a recognized, labeled adverse effect of statins, routinely monitored via liver function tests.

Case 05 – Levofloxacin (Tendon rupture)

Serious: Tendon rupture causes significant disability/incapacity and may require surgical intervention, meeting the seriousness criterion. Expected: Tendinitis and tendon rupture are subject to a boxed warning for the fluoroquinolone class and are therefore documented in the reference safety information.

Summary

Seriousness	Count	Cases
Serious	3	02, 03, 05
Non-Serious	2	01, 04

Out of 5 reviewed cases, 3 (60%) qualify as Serious ADRs requiring expedited regulatory reporting timelines (e.g., 15-day reporting to CDSCO/regulatory authority for serious cases), while 1 case (Metformin – lactic acidosis) is additionally Unexpected, warranting the most stringent reporting timeline and enhanced causality assessment.

— End of Report —